

JASON HON

SOPHMORE COMPUTER SCIENCE MAJOR

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⚙️ TECHNICAL SKILLS

Languages | Java, Python, C, C++, Javascript, Typescript, Scheme, Bash, CSS, HTML, SQL

Technologies | EC2, Nginx, Flask, Express, MongoDB, Node, React, Angular, Tailwind

Development | Git, Linux, Make, Valgrind, GDB, GCC

🎓 EDUCATION

University of Waterloo

Honours Bachelor of Computer Science, Artificial Intelligence Specialization

Waterloo, ON (Sept 2023 - 2027 Dec)

- GPA: 3.95 / 4.0 (CAV: 92.1%)
- Relevant Coursework: Data Structures, Object Oriented Programming, Compilers
- Honours: CS Intl. Upper Year Scholarship, Presidential Scholarship of Distinction (Totalled CAD\$ 8,000)

🏗️ PROJECTS

🔗 SLMod — Java, Flask, MongoDB

- Produced a portable **Java** QOL game modification used by over **40** users daily at its peak
- **Led a team of 3 developers** to develop a **Flask** user-tracking system, data stored using **MongoDB**
- Visualized **over 6,800** user log-ins on an online dashboard, implementing **OAuth2** for secure access.

🔗 Chess Engine — C++

- Developed a Chess engine in C++ featuring various levels of **AI opponents**, using advanced decision-making algorithms to incorporate piece values and to calculate complex evaluations
- Strictly adhered to **OOP** principles and advanced design patterns to create a structured and maintainable codebase, ensuring high cohesion and low coupling
- Led a team of 3 to maintain comprehensive documentation and UML diagrams to streamline development

🔗 UW Assignment Planner — MongoDB, Express, Angular, NodeJS

- Created an assignment planner for UWaterloo students using the **MEAN** stack
- Built **RESTful APIs** with **Express.js** for storing and updating tasks
- **Integrated third-party APIs** (e.g. Google calendar and notification) to enhance functionality

🔗 Boulder Solver — Java

- Developed a solver that solves complex boulder puzzles (Sokuban) in **Java** in requirements of a client
- Engineered a backtracking algorithm that generates the most optimized path for any maps in under **100ms**

🔗 Portfolio Website — React

- Developed a modern and responsive portfolio website using React, showcasing past experiences.
- Utilized React Router for seamless navigation between different sections of the website

🏢 EXPERIENCES

Bonumcare

Software Developer

Seattle, WA (Summer 2023)

- Engineered a "Conversation Buddy" for the elderly on an **Express** server, hosted on **AWS EC2** and deployed with **Nginx**, implementing APIs to send and receive **over 300** messages daily.
- Utilized **React**, **HTML**, **CSS** to redesign the existing interactive chat bot user interface, receiving positive feedback from 90% of elderly users
- Authored comprehensive documentation for the chatbot to facilitate future maintenance and code reuse

Content Creation

YouTube

(March 2022 - May 2023)

- Produced video content for a broad audience of an average of **60,000** viewers on YouTube
- Negotiated and finalized strategic partnerships with multiple sponsors (e.g. **Plarium**, **Akrew**)